

Vinh Ly

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Summary

Software Engineer with **18 years** of experience building and operating distributed systems at scale for Google, YouTube, and Microsoft. Led end-to-end development of services handling billions of requests daily, managed teams of 3-8 engineers, and drove technical strategy across product and infrastructure domains. Built systems on **Google Cloud Platform, Kubernetes, Bigtable**, and internal distributed frameworks. Collaborated with cross-functional partners to ship features reaching hundreds of millions of users. Focus on continuing to build reliable, maintainable systems that handle extreme scale. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: TypeScript, JavaScript, HTML, CSS

Distributed Systems: Kubernetes, Bigtable, Spanner, MapReduce, distributed tracing, load balancing

Cloud Infrastructure: Google Cloud Platform, containerization, auto-scaling, capacity planning, monitoring

Programming: Python, Java, C++, Go, SQL

Practices: on call, incident management, code review, mentorship, technical roadmaps, Docker, Postgres, AWS

Work Experience

Staff Software Engineer

YouTube | 02/2025 - Present

- Led backend infrastructure for video hosting and streaming at YouTube scale, designing systems for reliability and performance across multiple regions and service dependencies.
- Reduced video playback latency by implementing caching improvements and optimizing video delivery pipelines serving hundreds of millions of daily active users.
- Built automated capacity planning tools that predicted resource needs based on traffic patterns, reducing manual intervention during peak usage periods.
- Mentored team of 5 engineers on distributed systems design, performance debugging, and incident response practices for 24/7 on-call rotations.
- Collaborated with product partners to define technical requirements for new video features, coordinating cross-team launches and post-launch stability.

Staff Software Engineer

Whatnot | 05/2024 - 02/2025

- Led backend development for live commerce platform handling real-time bidding, checkout, and inventory systems serving thousands of concurrent users during peak events.
- Managed on-call rotation and incident response for platform reliability, coordinating with engineering teams during high-traffic drops and system degradations.
- Designed and executed migration of critical services from monolithic architecture to containerized deployment on Kubernetes, improving deployment frequency.
- Implemented distributed tracing across microservices to improve debugging capabilities, reducing mean time to resolution for production incidents.

Staff Software Engineer/Manager

Google | 03/2015 - 04/2024

- Drove technical direction for backend services supporting Google Search and core infrastructure products, setting architectural standards and technical roadmaps for 3-5 person teams.
- Improved reliability metrics for critical search serving systems, reducing error rates and improving response time consistency across distributed infrastructure.
- Built large-scale data processing pipelines using Bigtable, Spanner, and MapReduce to serve and analyze data across multiple Google products.
- Managed team of 8 engineers through hiring, performance reviews, and career development while maintaining delivery on infrastructure roadmaps.

Software Engineer

Google | 02/2011 - 02/2015

- Developed backend systems for Google Search infrastructure handling query processing and result ranking across distributed compute clusters.
- Collaborated with cross-functional teams on feature launches, operational readiness reviews, and production incident response for search infrastructure.
- Mentored junior engineers on systems design principles and code review practices within the search infrastructure engineering organization.
- Optimized data pipeline throughput for indexing systems, improving processing efficiency for large-scale web content analysis.

Software Design Engineer II

Microsoft | 10/2007 - 12/2011

- Developed software for Windows infrastructure and cloud services, building and testing features for enterprise-scale distributed systems.
- Participated in on-call rotations for production systems, triaging and resolving incidents affecting enterprise customer environments.
- Designed backend components for service-oriented architecture supporting Microsoft enterprise products and customer-facing deployments.

Software Design Engineer Intern

Microsoft | 06/2006 - 08/2006

- Contributed to development and testing of software components for Microsoft enterprise infrastructure platforms during summer internship program.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007